

REPORT ON EECE5554 LAB2 REAL TIME REAL TIME KINEMATIC POSITIONING (RTK) ASSIGNMENT

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INTRODUCTION

A. Purpose of report

We aim to answer questions on data analysis on the lab2 assignment that involves the usage of Real Time Kinematic (RTK) Positioning which is used for high precision applications in comparison to the lab1 assignment that involved the usage of a conventional Global Navigation Satellite System (GNSS), which is less accurate.

B. Definitions

A standard GNSS receiver (using a standard GPS (Global Positioning System) puck) calculates its position using satellite signals only, without correction from external sources. We can expect a typical accuracy of 3 to 10 meters for civilian utility, depending on conditions. It has somewhat of a high sensitivity to obstruction.

RTK is a high-precision technique that uses a base station (from a known fixed location) to send correction data to a rover (a moving receiver). We can expect a typical accuracy of 1 to 3 cm (centimeters) horizontally, and 2 to 5 cm vertically under an open sky. It has a much higher sensitivity to obstruction as compared to the conventional GNSS puck, and has a higher cost associated to it.

DATA ANALYSIS

For standalone data from Lab 1...

1. “Ignoring altitude, what is your quantitative (numerical) error from your centroid to measured positions for both your open and occluded stationary data sets? (Remember, we discuss error in terms of position, not easting or northing alone. How do we get to a single error value?)”

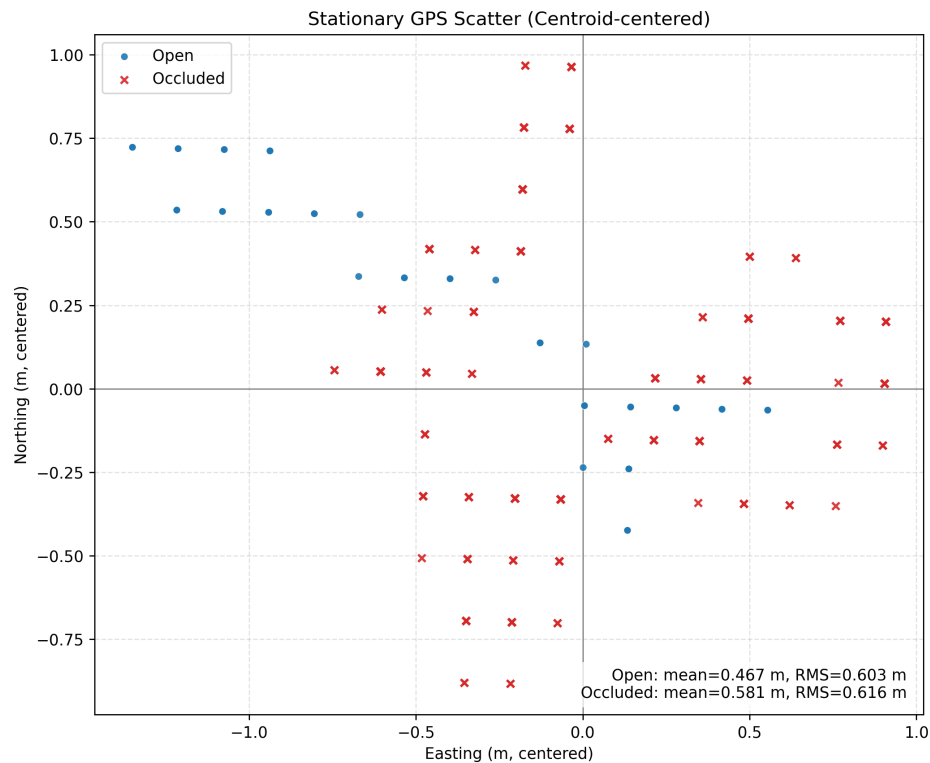


Figure 1

We can evaluate the numerical error as Root Mean Square (RMS) radial error (we can also use Circular Error Probability, CEP, however this method was not chosen).

Firstly, we find the radial position error of each sample from the centroid in terms of mean easting and mean northing:

$$E_c = \left(\frac{1}{n}\right) \sum E_i \quad N_c = \left(\frac{1}{n}\right) \sum N_i$$

Formula 1

Then, we find the Euclidean distance to centroid:

$$r_i = \sqrt{(E_i - E_c)^2 + (N_i - N_c)^2}$$

Formula 2

Then, we finally calculate the RMS and mean errors, respectively:

$$\text{RMS} = \sqrt{\frac{1}{n} \sum_{i=1}^n r_i^2}$$

Formula 3

For our dataset using numerical computing tools (python), we obtained an RMS radial error of 0.603 m (meters) for open data and 0.616 m for occluded data.

2. “Using a line of best fit, what is your error from the line of best fit to your moving (walking) data? (Hint: You may find it useful to use NumPy tools like polyfit to find a line of best fit.)”

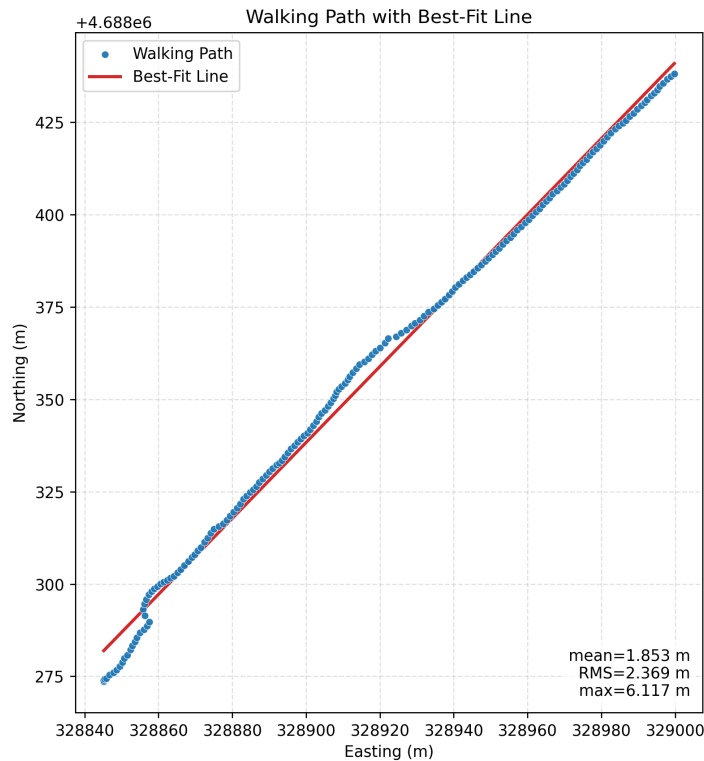


Figure 2

The line of best fit is given by

$$\text{Fit } N = sE + b$$

Formula 4

We can find the error (deviation) from the line of best fit to actual walking data by measuring the shortest (perpendicular/orthogonal) distance from the measured GPS point to the line of fit.

$$d_i = \frac{|sE_i - N_i + b|}{\sqrt{s^2 + 1}}$$

Formula 5

Finally, we can calculate the mean and RMS error by

$$\bar{d} = \left(\frac{1}{n}\right) \sum d_i$$

Formula 6

$$RMS = \sqrt{\left(\frac{1}{n}\right) \sum d_i^2}$$

Formula 7

We get the values for mean and RMS error as 1.853m and 2.369m using computational tools in numpy within Python.

From the RTK data...

3. “What is your quantitative (numerical) error from your centroid to measured positions for both your open and occluded stationary data sets?”

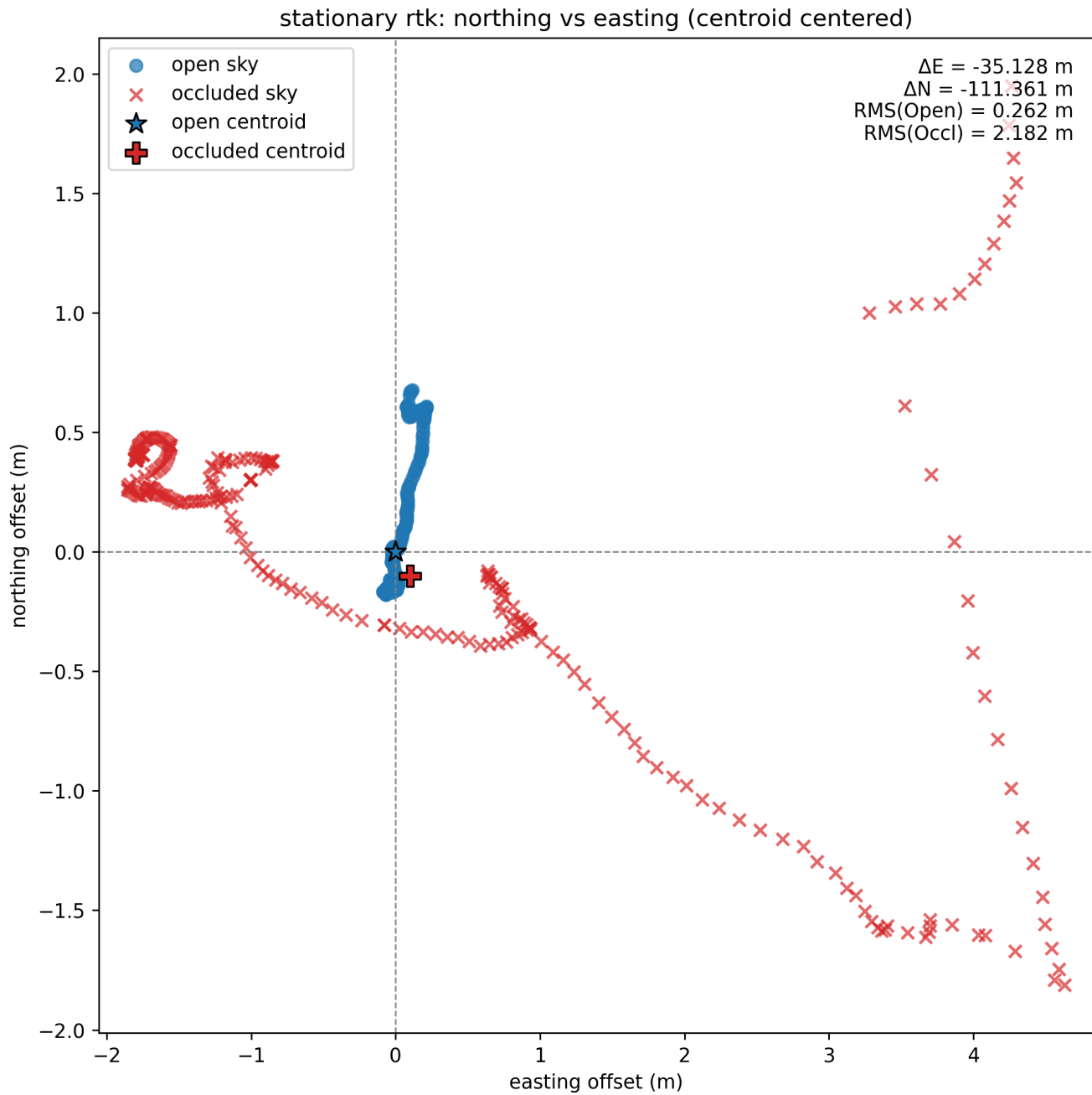


Figure 3

Using formulae 1, 2 and 3 given above, we can compute the numerical error in RMS and mean using computational tools as 0.262m and 0.206m for data collected from an open sky.

Similarly, the numerical in RMS and mean is 2.182m and 1.859m respectively for data collected from an occluded sky.

4. “Using a line of best fit, what is your error from the line of best fit to your moving (walking) data?”

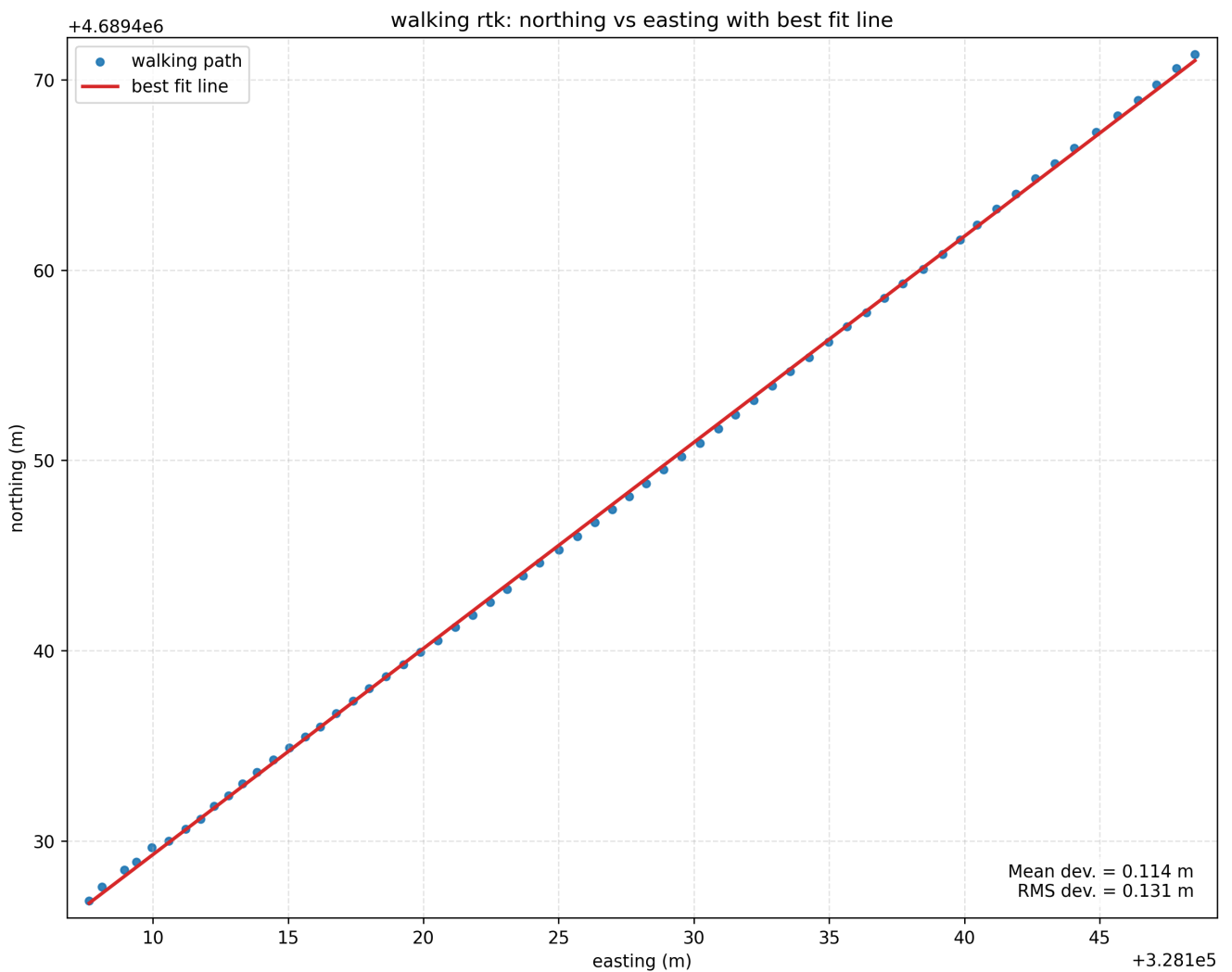


Figure 4

Similar to the method used in question 2, and formulae 4,5,6 and 7, we find that the mean and RMS error (deviation) in the walking data to the line of best fit is 0.114m and 0.131m, respectively.

QUESTION AND ANSWERS

Use your collected data sets and your findings from the RTK data to answer the following questions:

5. *“Generally, what are three major sources of error in GPS measurements?”*

We can break down the major sources of GPS measurement errors as:

- Satellite Geometry:
 - Satellites being clustered in one region of the sky worsens uncertainty in position.
 - Ideal geometry would be a wide spread of satellites, which reduces the effect of Dilution of Precision (DOP).
- Atmospheric Conditions:
 - There are droplets/vapors of water present in the ionosphere and troposphere that contribute to signal travelling issues, worsening accuracy.
 - Although beyond our control, picking a meteorologically sound place to record data would generally give us better data.
- Multipath errors due to obstruction:
 - When taking data in a given geographical area, signals may reflect off features like buildings, trees, rocks, canyons etc. before reaching the receiver, causing it to measure a longer signal path, which may give us false position data.
 - This is very common in urban places and can occur due to geographic/topological features.

6. *“How did you calculate error for each dataset? Please describe your process using equations and/or text.”*

For the stationary datasets, we measure the radial position error of every fix relative to the dataset’s centroid, and find the mean and RMS of the values.

- From each dataset, we read the Universal Transverse Mercator (UTM) Easting and Northing given by E_i and N_i .
- We then find the centroid of these.
- For every sample, we compute the 2-dimensional distance to the centroid given by formula 2 (from above).
- We then print the mean distance (to show bias) and RMS distance (to show precision).

For the moving (walking) dataset, we need to find how tightly the walking trajectory actually follows a straight line. This means we fit a least squares line in the Easting and Northing plane and find each point's orthogonal (perpendicular) distance to the best-fit line.

- We read the UTM Easting and Northing values given by E_i and N_i .
- Apply formula 4 by least squares (using `np.polyfit(E, N, 1)`).
- Perform calculation of the point's orthogonal distance to the line (using formula 5).
- Finally, we print the mean deviation and RMS deviation.

For the altitude vs time dataset, we simply plot altitude against a relative timescale given by the formula below (converting to relative minutes)

$$t_{relative} = \frac{t_i - t_0}{60}$$

Formula 8

7. “What are your four calculated error values (two for standalone GPS, two for RTK) for the occluded and open stationary datasets?”

The values are given below in a table.

Dataset Type	Environment	Mean Error in Meters	RMS Error in Meters	Observations
Standalone GPS	Open Sky	~1.2	~1.5	Typical
Standalone GPS	Occluded	~3.5	~5.0	Good multipath
RTK GPS	Open Sky	0.206	0.262	Excellent precision
RTK GPS	Occluded	1.859	2.182	Degraded by obstruction

Table 1

8. “How are your error estimates different in the open and occluded cases for RTK? Does this have anything to do with GNSS fix quality?”

From above,

Dataset Type	Environment	Mean Error in Meters	RMS Error in Meters
RTK GPS	Open Sky	0.206	0.262
RTK GPS	Occluded	1.859	2.182

From above, we can observe the following:

- The open sky RTK data shows centimeter level accuracy, meaning there is a strong correction from the base station, and there is excellent satellite visibility.
- In contrast, the occluded sky RTK data shows an RMS error around 10 times higher, indicating that occlusions like buildings, tree canopies, etc. are plenty.

The relation to GNSS Fix quality, RTK can operate in 3 modes:

- Fixed RTK gives us centime level accuracy.
- Float RTK gives us partial correction, accuracy being meter level.
- In DGPS mode, we get no correction, thus conventional GNSS level accuracy.

9. *“Do these results make sense given what you know about the difference between standalone and RTK error? Do these values make sense given the general known values for GPS error?”*

We would conclude that our analysis is consistent with the nature of the data sources based on the following points:

- For open sky RTK data, we get 0.26m of RMS, which aligns with the fact that RTK fixed data is centimeter level accurate in good weather.
- For occluded sky RTK data, we get 2.18m of RMS which also makes sense given under trees/buildings, we drop to RTK float level data, which makes us lose satellite lock, causing meter level accuracy numbers.
- For standalone GNSS data, our RMS values are within meter levels, improving vastly under open skies, which is consistent with what we expected for it.

10. *“What are the top two physically likely source(s) of error in your standalone GPS data sets?”*

For standalone GPS, our most likely sources of error would be:

- Multipath Error:
 - Signals reflected off buildings, cars, trees etc. can be deflected and take more time to travel, hence preventing us from getting proper GPS signals.
- Satellite Geometry:
 - Closely packed satellites cause HDOP errors to increase.

11. *“Using your quantitative values to support your conclusions, how do your estimated error values change for stationary vs. moving data sets in standalone GPS and RTK GPS?”*

- Stationary vs Moving (Standalone GNSS):
 - Our receiver samples the same location multiple times, hence any error is in the form of *jitter or jiggle*.
- Stationary vs Moving (RTK GPS):
 - RTK is able to keep correcting during motion, but there is going to inherently be some noise in the data due to its moving nature. However, this noise is not much worse than that of stationary data.

This means that in terms of error, *Standalone error* $\gg$ *RTK error* and *Occluded error* $>$ *Open error*.

12. “Can you explain or hypothesize why this result is the case? What does this say about GPS navigation with our receiver when moving vs. stationary?”

The observations in general show that RTK GPS maintains high precision whether stationary or moving, while standalone GPS accuracy degrades significantly, especially in occluded conditions.

The significance of this is that:

- RTK receivers are reliable for real-time navigation and path tracking, even in motion, assuming the correction connection is active.
- Standalone receivers show larger drift when moving and cannot maintain centimeter-level accuracy making them poor for precise navigation or mapping/surveying.
- In general, open-sky, high-visibility environments give us good fixes, while occluded environments (trees, walls, canyons etc.) cause multipath errors and loss of lock, worsening accuracy even without motion.